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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IACUC: Materials and methods Experimental animals and middle cerebral artery occlusion (MCAO) model All animal experiments were performed with the approval and in accordance with standard principles approved by the Institutional Animal Care and Use Committee of China Resources & WISCO General Hospital (approval no. 2020-005 Hubei, China).
Euthanasia Agents: After anesthesia with pentobarbital sodium (45 mgkg), a monofilament nylon suture was inserted into the internal carotid artery through the external carotid artery, followed by occluding the origin of the MCA.
Inclusion and Exclusion Criteria
Mice that died or failed to show an 80 reduction in cerebral blood flow after occlusion were excluded from the following experiments.
Attrition
not detected.
Sex as a biological variable
The Experimental Animal Center of the Chinese Academy of Medical Sciences (Beijing, China) provided 90 male C57BL6J mice (age, 8-10 weeks body weight, 22-30 g) for the current experiments.
Subject Demographics
Age: The Experimental Animal Center of the Chinese Academy of Medical Sciences (Beijing, China) provided 90 male C57BL6J mice (age, 8-10 weeks body weight, 22-30 g) for the current experiments.
Weight: The Experimental Animal Center of the Chinese Academy of Medical Sciences (Beijing, China) provided 90 male C57BL6J mice (age, 8-10 weeks body weight, 22-30 g) for the current experiments.
Randomization
In total, five wells were randomly selected for each treatment group.
Blinding
Neurological scores were assessed at 24 , 48 and 72 h after MCAO by technicians blinded to the treatment groups .

Power Analysis
not detected.
Replication
Western blot analysis Western blot analysis was repeated three times in this study.
Number: Western blot analysis Western blot analysis was repeated three times in this study.
Cell Line Authentication
not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
After the membranes were blocked with 5 skimmed milk for 2 h at room temperature, they were cultured with anti-FOXO3 (cat. no. ab70315 1:2,000 Abcam) and anti-GAPDH (cat. no. ab9485 1:2,500 Abcam) antibodies overnight at 4degC.	anti-FOXO3		
	anti-GAPDH		
Then, the membrane was incubated with a goat anti-rabbit HRP-conjugated secondary antibody (cat. no. ab205718 1:2,000 Abcam) at room temperature for 1 h.	anti-rabbit		
Experimental Models: Cell Lines			
N2a cells (2x107) after indicated transfection were cross-linked with 0.75 formaldehyde.	N2a		
Experimental Models: Organisms/Strains			
The Experimental Animal Center of the Chinese Academy of Medical Sciences (Beijing, China) provided 90 male C57BL6J mice (age, 8-10 weeks body weight, 22-30 g) for the current experiments.	C57BL6J		Suggestion: RRID:IMSR_JAX:000664(link)
The mouse neuroblastoma cell line N2a was obtained from the American Type Culture Collection , and was cultured in DMEM (Gibco Thermo Fisher Scientific , Inc .) containing 10 FBS (Gibco Thermo Fisher Scientific , Inc .) , 1 penicillinstreptomycin (Gibco Thermo Fisher Scientific , Inc .) at 37degC with 5 CO2 .	N2a		
Recombinant DNA			
Ltd. miR-27a-3p mimics and inhibitors , as well as their corresponding NCs (NC mimics , NC inhibitor at a concentration of 5 uM) , pcDNA3.1FOXO3 , pcDNA3.1 and pcDNA3.1XIST were purchased from Shanghai GenePharma Co . , Ltd.	pcDNA3.1		Suggestion: RRID:Addgene_100544(link)
miR-27a-3p mimics and inhibitor , NC mimics , NC inhibitor , pcDNA3.1FOXO3 , pcDNA3.1 and pcDNA3.1XIST are provided in Tables I-III .	pcDNA3.1FOXO3		
	pcDNA3.1XIST		

Luciferase reporter assay The sequences of XIST or FOXO3 were inserted into the luciferase reporter vector pmirGLO (Promega Corporation) to construct pmirGLO-XIST-wild-type (WT) or pmirGLO-FOXO3-WT reporters.	pmirGLO		
	pmirGLO-XIST-wild-type		
	pmirGLO-FOXO3-WT		
The reporters of pmirGLO-XIST-mutant (MUT) or pmirGLO-FOXO3-MUT were generated by site-directed mutagenesis.	pmirGLO-FOXO3-MUT		
Software and Algorithms			
Stained images were captured with a camera , and the infarct area was measured with ImageJ version 6.0 software (National Institutes of Health) .	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
The target genes of miR-27a-3p were identified using the TargetScan database (version 7.2 http:www.targetscan.org).	TargetScan		
The data of flow cytometry were analyzed using Flowing version 2.5.1 software (Turku Bioscience) and Origin version 8 software (OriginLab).	Flowing		
	Origin		
The Quantity One software (4.5.0 basic Bio-Rad Laboratories, Inc.) was used for the semi-quantification of protein expression.	Quantity One		
	Bio-Rad Laboratories		Suggestion: Bio-Rad Laboratories (RRID:SCR_008426)(link)
Statistical analysis Statistical analysis was performed with GraphPad Prism 5 software (GraphPad Software, Inc.).	GraphPad Prism		Suggestion: GraphPad Prism (RRID:SCR_002798)(link)
	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Statistical significance between two different groups was evaluated using an unpaired Students t-tests, while differences among >2 groups were determined according to one-way ANOVA followed by Tukeys post hoc test.	t-tests
	ANOVA
	Tukeys post hoc test
Oligonucleotides	
miR-27a-3p F , 5-TTCACAGTGGCTAAGTTCCGC- 3 and R , 5-CTCTACAGCTATATTGCCAGCCAC-3 FOXO3 F , 5-ATCTACGAGTGGATGGTGC-3 and R , 5-CCGGATGGAGTTCTTCCAG-3 GAPDH F , 5-GATCATCAGCAATGCCTCC-3 and R , 5-TCCACGATACCAAAGTTGTC-3 and U6 F , 5-CAATACAGAGAAAGTTAGCACG-3 and R , 5-AATGCTTCAAAGAGTTGTGC-3 .	5-CCGGATGGAGTTCTTCCAG-3 GAPDH F ,

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq